

ASSIGNMENT – 1

Software Engineering (UCS3504) – Hospital Management System: Comprehensive Analysis, SDLC, SRS, Use Case and DFD

AIM

To apply software engineering principles to analyze, model, and design a real-world software system. The chosen system is a Hospital Management System (HMS). This report presents a comprehensive study including problem definition, scope, objectives, risk analysis, ethics and sustainability, SDLC selection and schedule, feasibility study, requirements engineering with a full IEEE-style SRS, use case modeling with detailed scenarios, and data-flow design with a data dictionary.

1. INTRODUCTION AND PROBLEM DEFINITION

Modern hospitals operate in a high-pressure environment where accurate, real-time information can be the difference between efficient patient care and operational chaos. Many hospitals still rely on a mix of paper files, spreadsheets, and disconnected departmental applications. This fragmentation causes delayed appointments, duplicated tests, prescription errors, billing disputes, and poor resource utilization. The proposed Hospital Management System (HMS) is a unified, web-based platform that integrates patient registration, appointment scheduling, electronic medical records (EMR), diagnosis, prescriptions, pharmacy, laboratory, billing, admission/discharge, staff scheduling, inventory, and analytics.

- Target users: Patients, doctors, nurses, receptionists, pharmacists, lab technicians, billing staff, and hospital administrators.
- Primary stakeholders: Hospital management, clinical staff, patients and their families, insurance companies, pharmacy suppliers, medical device vendors, and government health authorities.
- Project scope: A multi-tenant, role-based web application supporting outpatient and inpatient workflows for a 250-bed multi-specialty hospital.
- Major modules: Patient Registration, Appointment Scheduling, EMR, Doctor Workbench, Pharmacy & Inventory, Laboratory, Billing & Payments, Admission/Discharge, Staff Scheduling, Reports & Analytics.
- Expected outcomes: Reduced average waiting time by 40%, elimination of duplicate records, faster billing cycles, centralized audit-ready data, and improved regulatory compliance.

1.1 PROJECT OBJECTIVES

Objective	Measurable target
Reduce patient wait time	Average OPD wait time under 15 minutes
Improve data accuracy	Less than 1% record duplication after first year

Objective	Measurable target
Faster billing	80% of bills generated within 5 minutes of service
Resource optimization	Bed occupancy visibility and staff scheduling improved by 25%
Audit compliance	100% traceability of prescription and billing changes

1.2 RISK ANALYSIS

Risk	Impact	Probability	Mitigation strategy	Owner
Data breach / unauthorized PHI access	High	Medium	RBAC, AES-256 encryption, TLS 1.3, audit logs, periodic security audits	Security team
System downtime during peak hours	High	Medium	Cloud HA, daily backups, failover, disaster recovery drills	Infrastructure team
Staff resistance to digital workflows	Medium	High	Phased rollout, hands-on training, super-user program, feedback loops	Change manager
Integration failure with lab devices	Medium	Medium	HL7/FHIR interfaces, test environment, middleware adapters	Integration team
Budget or schedule overrun	High	Low	Agile/incremental delivery, MVP-first, fixed-scope sprints	Project manager
Medication errors from data entry	High	Medium	Barcode scanning, drug-allergy alerts, double-check prompts	Clinical safety team

Risk	Impact	Probability	Mitigation strategy	Owner
Vendor lock-in to cloud provider	Medium	Low	Containerized deployment, portable database schema	Architecture team

2. ETHICS, LEGAL AND SUSTAINABILITY CONSIDERATIONS

Building an HMS involves sensitive personal health information. Ethical design must be baked into the system from the start, not added later.

- Patient privacy and confidentiality: The system must comply with the Digital Information Security in Healthcare Act (DISHA) India and follow HIPAA-style principles for data minimization, consent, and breach notification.
- Informed consent: Patients must explicitly consent to data sharing with insurers, researchers, or third-party providers; consent must be revocable.
- Data minimization: Collect only the data needed for care, billing, or compliance; avoid unnecessary personal identifiers.
- Fair access and digital inclusion: The UI should support local languages, high-contrast modes, and simple navigation so elderly and differently-abled users are not excluded.
- Algorithmic transparency: Any decision-support or triage scoring must be explainable and auditable, with human oversight.
- Sustainability: Cloud deployment reduces physical hardware and paper consumption, supporting SDG 9 (Industry, Innovation and Infrastructure) and SDG 12 (Responsible Consumption and Production).
- Professional responsibility: Developers, doctors, and administrators must follow ACM/IEEE code of ethics and not bypass safety checks for speed or profit.

3. SDLC MODEL SELECTION

The Incremental Software Development Life Cycle model is selected for the HMS. A hospital cannot pause operations for a big-bang replacement. The Incremental model lets the team build and deploy the system in prioritized, working increments, collect real user feedback, and extend the product over time. Each increment delivers a usable subsystem, reducing both technical and business risk.

3.1 JUSTIFICATION FOR INCREMENTAL MODEL

Factor	Why Incremental fits
Business continuity	Modules go live one at a time; old processes continue for unaffected departments
Risk reduction	High-risk modules (e.g., EMR, billing) are prototyped and validated early

Factor	Why Incremental fits
User feedback	Doctors and staff can use real increments and request changes before later modules are built
Budget control	Management can pause or reprioritize increments based on ROI
Regulatory testing	Each increment is validated independently, making audits easier

3.2 INCREMENTAL PLAN

Increment	Modules	Deliverable
Increment 1	Patient Registration + Appointment Scheduling	Working appointment portal and reception dashboard
Increment 2	EMR + Doctor Diagnosis/Prescription	Electronic records and prescription workflow
Increment 3	Pharmacy + Laboratory + Inventory	Medicine and test order management
Increment 4	Billing + Admission/Discharge + Reports	End-to-end revenue and administration workflow
Increment 5	Admin + Analytics + Integration	Hospital-wide dashboards and third-party integrations

3.3 PROJECT SCHEDULE

The proposed schedule covers 20 weeks, with planning in the first month and each implementation increment taking approximately three weeks.

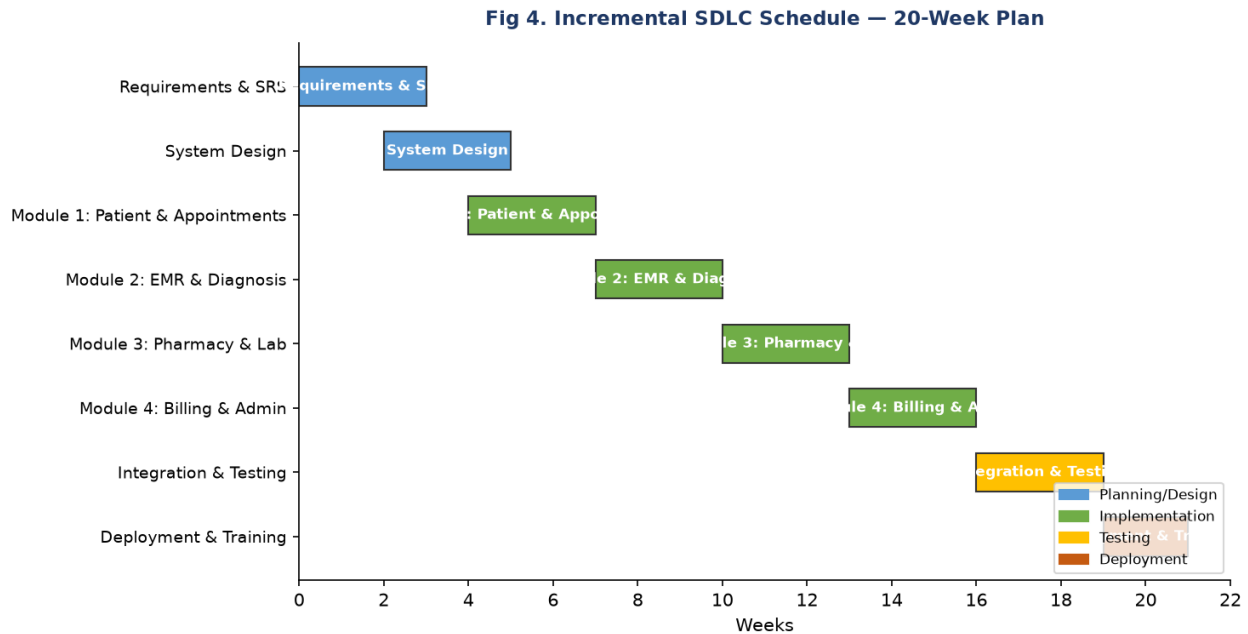


Fig 4. Incremental SDLC Gantt chart

4. FEASIBILITY ANALYSIS

A feasibility study evaluates whether the HMS project is worth pursuing. The analysis covers technical, economic, operational, legal, and schedule dimensions.

4.1 TECHNICAL FEASIBILITY

The system can be built using mature open-source and commercial technologies. The recommended stack is Spring Boot or Django for the backend, React for the frontend, PostgreSQL for transactional data, Redis for caching, and AWS/Azure/ on-premise private cloud for hosting. HL7/FHIR libraries and DICOM viewers are available for integration with lab and imaging systems. The hospital already has LAN/WAN connectivity, so no major network upgrade is needed.

4.2 ECONOMIC FEASIBILITY

Cost / Benefit item	Estimate
Initial development cost (5 increments)	₹ 25–30 lakhs
Annual cloud and maintenance cost	₹ 4–5 lakhs
Annual savings from reduced paper, faster billing, fewer duplicate tests	₹ 8–10 lakhs
Payback period	Approximately 3 years
Break-even point	After second year of operation

4.3 OPERATIONAL FEASIBILITY

The hospital has an IT helpdesk and network infrastructure. Doctors and nurses are familiar with computers, but a structured change-management program is essential. Operational feasibility is high because the system mirrors existing workflows rather than forcing a complete redesign.

4.4 LEGAL AND REGULATORY FEASIBILITY

The HMS must comply with the IT Act 2000, DISHA (India), and data-localization guidelines. With role-based access, encryption, audit logs, and consent management, compliance is achievable. A legal review of insurance and billing integration is recommended before go-live.

4.5 SCHEDULE FEASIBILITY

A 20-week incremental plan with a working MVP after 8 weeks is realistic for a team of 6–8 developers. Buffer time is included for user acceptance testing and integration.

4.6 FEASIBILITY SUMMARY

Feasibility type	Feasible?	Reason
Technical	Yes	Mature stack, existing network, reusable healthcare standards
Economic	Yes	Payback in ~3 years through efficiency gains
Operational	Yes	Existing IT support; workflows map to current practice
Legal	Yes with controls	Compliance achievable with RBAC, encryption, audit logs
Schedule	Yes	20-week incremental plan with MVP at week 8

5. REQUIREMENTS ENGINEERING

Requirements engineering identifies what the HMS must do and how well it must do it. Functional requirements define behavior; non-functional requirements define quality attributes.

5.1 FUNCTIONAL REQUIREMENTS

ID	Requirement	Priority
FR-01	The system shall register a new patient, generate a unique patient ID, and store demographics, contact, insurance, and emergency details.	High
FR-02	The system shall allow patients or receptionists to search doctors by department and book, reschedule, or cancel appointments without double-booking.	High
FR-03	The system shall let doctors view patient history, add diagnosis notes, prescribe medicines, and order lab tests.	High
FR-04	The system shall dispatch prescriptions to the pharmacy and update the dispensing status.	High
FR-05	The system shall generate consolidated bills from consultation, lab, pharmacy, and room charges, and support cash, card, UPI, and insurance payments.	High
FR-06	The system shall manage admission, bed allocation, and discharge with discharge summaries.	Medium
FR-07	The system shall maintain medicine and consumable inventory with stock alerts.	Medium
FR-08	The system shall schedule doctors and nursing staff by department and shift.	Medium
FR-09	The system shall provide dashboards and reports for administration.	Medium
FR-10	The system shall notify patients and staff via SMS/email for appointments, reports, and bills.	Low

5.2 NON-FUNCTIONAL REQUIREMENTS

ID	Requirement
NFR-01	Availability: 99.9% uptime during operating hours; critical modules must failover within 60 seconds.
NFR-02	Performance: Search and booking response time under 2 seconds for 95% of requests.
NFR-03	Security: AES-256 at rest, TLS 1.3 in transit, RBAC, MFA for admins, and immutable audit logs.
NFR-04	Scalability: Support 100 concurrent users in phase 1 and 1,000 users by year 3.
NFR-05	Usability: Responsive UI, multi-language support, accessibility compliance (WCAG 2.1 Level AA).
NFR-06	Reliability: Daily automated backups, point-in-time recovery within 24 hours.
NFR-07	Maintainability: Modular architecture, test coverage above 80%, and CI/CD pipelines.

5.3 IEEE SRS DOCUMENT OUTLINE

Section	Content summary
1. Introduction	Purpose, scope, definitions, acronyms, references, overview
2. Overall Description	Product perspective, functions, user classes, operating environment, constraints
3. System Features	All FRs with priority, stimuli/response sequences, related use cases
4. External Interface Requirements	User interfaces, hardware interfaces, software interfaces, communication
5. Non-Functional Requirements	Performance, safety, security, availability, maintainability, portability
6. Data Requirements	Logical data model, entity-relationship diagram, data dictionary
7. Appendices	Assumptions, dependencies, acceptance criteria, traceability matrix

6. USE CASE MODEL

The use case diagram identifies the actors and the major functions they perform. Patients interact with registration and appointments; doctors with diagnosis and prescriptions; receptionists with admissions; pharmacists, lab staff, billing staff, and administrators with their respective modules.

Fig 1. Use Case Diagram — Hospital Management System

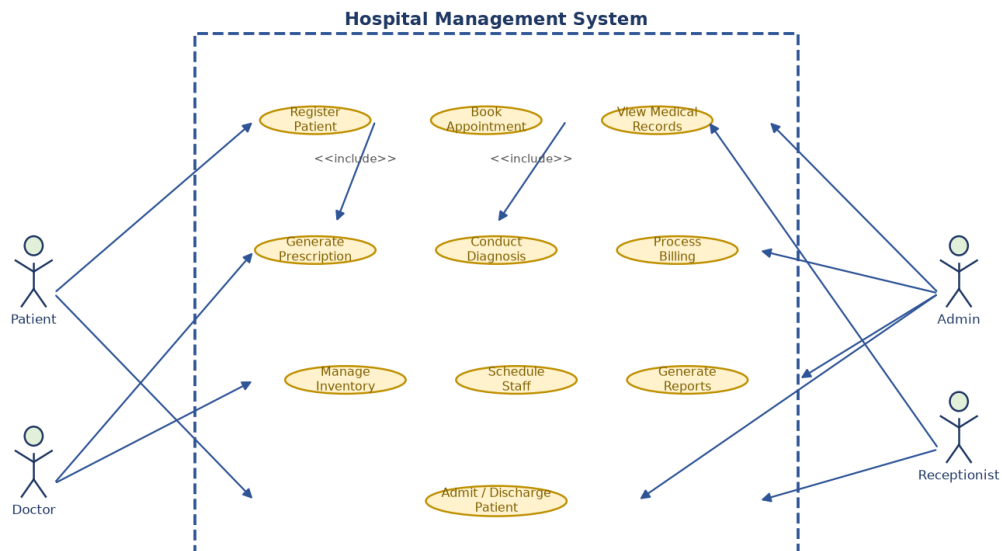


Fig 1. Use Case Diagram — Hospital Management System

6.1 USE CASE DESCRIPTION: BOOK APPOINTMENT

Field	Description
Use case ID	UC-01
Use case name	Book Appointment
Actors	Patient, Receptionist
Description	A patient or receptionist searches a doctor by department or name, selects an available time slot, and confirms an appointment.
Precondition	Patient is registered; doctor schedule is configured.
Main flow	1. Actor selects department. 2. System lists doctors. 3. Actor chooses a doctor. 4. System shows available slots. 5. Actor selects slot. 6. System confirms booking. 7. System sends notification.
Alternate flow	A1. No slot available: system suggests next available date/time.

Field	Description
Postcondition	Appointment record created, doctor schedule updated, patient notified.

6.2 USE CASE DESCRIPTION: CONSULT PATIENT

Field	Description
Use case ID	UC-02
Use case name	Consult Patient
Actors	Doctor
Description	Doctor reviews the patient's medical history, records the current diagnosis, writes prescriptions, and orders lab tests.
Precondition	Doctor is logged in; patient has an active appointment.
Main flow	1. Doctor opens patient record. 2. System displays history and vitals. 3. Doctor enters diagnosis. 4. Doctor selects medicines from formulary. 5. Doctor sets dosage and duration. 6. Doctor orders lab tests if needed. 7. System saves EMR and notifies pharmacy/lab.
Alternate flow	A1. Medicine unavailable: doctor selects an alternative. A2. Allergy alert: system warns doctor of known allergy.
Postcondition	Consultation record and prescription stored; pharmacy and lab receive requests.

6.3 USE CASE DESCRIPTION: PROCESS BILLING

Field	Description
Use case ID	UC-03
Use case name	Process Billing
Actors	Billing Staff
Description	Billing staff retrieves the patient's service history, applies insurance or discount rules, generates an invoice, and records payment.
Precondition	Patient services are completed and charges are posted from consultation, lab, and pharmacy.

Field	Description
Main flow	1. Staff selects patient. 2. System aggregates charges. 3. Staff applies insurance/discount. 4. System generates invoice. 5. Staff receives payment. 6. System prints/email receipt.
Alternate flow	A1. Partial payment: system records balance and generates payment plan.
Postcondition	Payment recorded, accounts updated, receipt issued.

6.4 USE CASE DESCRIPTION: MANAGE INVENTORY

Field	Description
Use case ID	UC-04
Use case name	Manage Inventory
Actors	Pharmacist, Admin
Description	Users update medicine and consumable stock, set reorder levels, and generate purchase requests.
Precondition	User has inventory-management role.
Main flow	1. User views current stock. 2. System flags items below reorder level. 3. User updates quantities or creates purchase order. 4. System logs transaction.
Alternate flow	A1. Item out of stock: system blocks prescription dispensing until restocked.
Postcondition	Inventory records updated; purchase order generated if needed.

7. DATA FLOW MODEL

The Data Flow Diagram (DFD) shows how data moves between external entities, processes, and data stores. The Context DFD presents the HMS as a single process. The Level 1 DFD decomposes it into nine interacting processes with six data stores and the relevant external entities.

7.1 CONTEXT DIAGRAM

Fig 2. Context Diagram (Level 0 DFD)

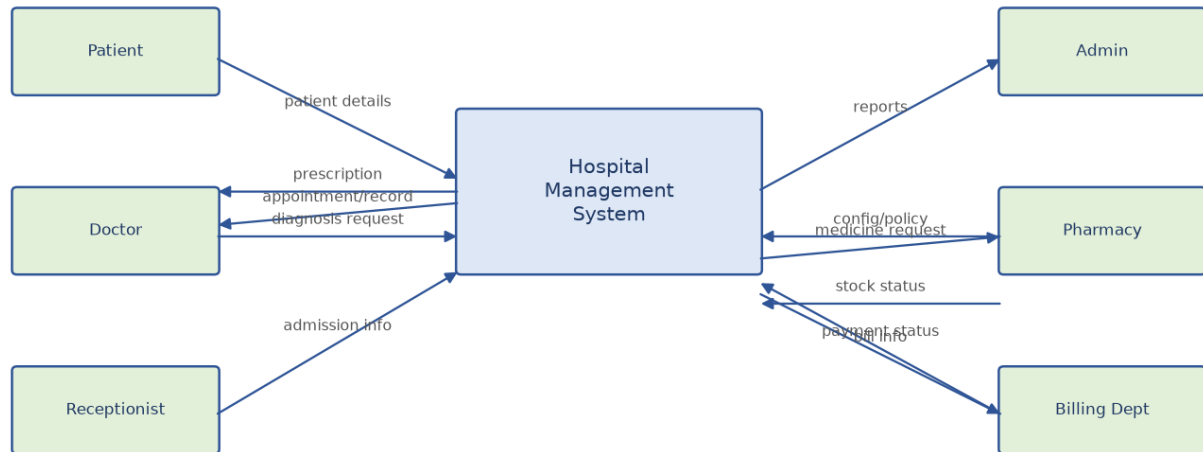


Fig 2. Context Diagram (Level 0 DFD)

7.2 LEVEL 1 DFD

Fig 3. Level 1 Data Flow Diagram — Hospital Management System

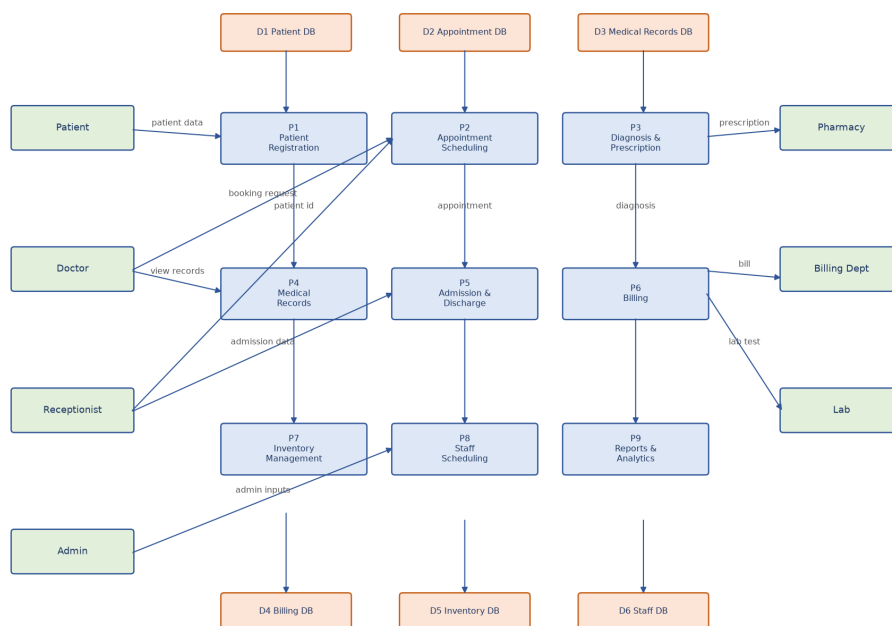


Fig 3. Level 1 Data Flow Diagram

7.3 DFD DESCRIPTION

Process	Input	Output	Data stores
P1 Patient Registration	Patient data	Patient ID	D1 Patient DB

Process	Input	Output	Data stores
P2 Appointment Scheduling	Booking request	Appointment confirmation	D2 Appointment DB
P3 Diagnosis & Prescription	Symptoms, history	Diagnosis, prescription	D3 Medical Records DB
P4 Medical Records	Patient ID	Medical history	D3 Medical Records DB
P5 Admission & Discharge	Admission data	Bed status, discharge summary	D1 Patient DB, D4 Billing DB
P6 Billing	Service charges	Invoice, receipt	D4 Billing DB
P7 Inventory Management	Stock update, usage	Stock alert, purchase order	D5 Inventory DB
P8 Staff Scheduling	Admin inputs	Schedule, leave status	D6 Staff DB
P9 Reports & Analytics	Aggregated data	Dashboard, reports	All data stores

7.4 DATA DICTIONARY

Data store / flow	Description	Key data elements
D1 Patient DB	Master patient demographics and contact/insurance info	patient_id, name, DOB, gender, phone, address, insurance_id
D2 Appointment DB	Appointment slots, status, and history	appointment_id, patient_id, doctor_id, date, time, status
D3 Medical Records DB	EMR including diagnosis, prescriptions, lab results	record_id, patient_id, diagnosis, prescription, notes, attachments
D4 Billing DB	Invoices, payments, dues, insurance claims	bill_id, patient_id, items, total, payment_status
D5 Inventory DB	Medicine and consumable stock levels	item_id, name, quantity, expiry, reorder_level
D6 Staff DB	Doctors, nurses, and admin staff schedules	staff_id, name, role, department, shift, leave
Patient data	Flow from patient/receptionist during registration	demographic and contact details
Prescription	Flow from doctor to pharmacy	medicine, dosage, duration

Data store / flow	Description	Key data elements
Bill info	Flow from billing process to payment	charges, discounts, total

8. TESTING AND QUALITY ASSURANCE

- Unit testing: Each service and controller tested with JUnit/Mocha; target 80% code coverage.
- Integration testing: Verify HL7/FHIR message flow between EMR, lab, and pharmacy modules.
- System testing: End-to-end workflows including patient registration, appointment, consultation, billing, and discharge.
- User acceptance testing: Doctors, nurses, and receptionists validate real-world scenarios in a staging environment.
- Security testing: Penetration testing, RBAC verification, and audit-log completeness.

9. ASSUMPTIONS AND DEPENDENCIES

- The hospital will provide accurate master data (doctors, departments, room inventory, medicine catalog) before go-live.
- Stable internet and LAN/WAN connectivity are available at all hospital locations.
- Legacy paper records will be digitized for active patients only; historical records can be scanned on demand.
- Insurance company APIs or file formats for claim submission are available.
- Regulatory requirements (DISHA, IT Act) remain stable during development.

10. CONCLUSION

The proposed Hospital Management System addresses the major operational pain points of a multi-specialty hospital through an integrated, incremental, and secure software solution. The analysis shows that the project is technically, economically, operationally, legally, and schedule-feasible. The Incremental SDLC model is appropriate because it allows the hospital to adopt modules gradually without disrupting daily operations. Use case and DFD models confirm that patient data flows cleanly through registration, appointment, consultation, pharmacy, lab, billing, and administration. With strong security, privacy, and sustainability practices, the HMS can improve patient care quality, reduce administrative overhead, and support regulatory compliance.

11. LEARNING OUTCOMES

- Defined a real-world problem statement, scope, stakeholders, measurable objectives, and risk mitigation for the HMS.
- Analyzed ethical, legal, and sustainability concerns and justified the Incremental SDLC with a detailed schedule.
- Performed comprehensive technical, economic, operational, legal, and schedule feasibility analysis.
- Specified functional and non-functional requirements and produced an IEEE-style SRS outline.
- Developed a use case model with four detailed use-case descriptions covering key HMS workflows.
- Designed Context and Level 1 DFDs with a supporting data dictionary and process descriptions.